SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5-AT3 Hackathon**	**Assessment:**	**ASSESSMENT TOOL 3 – Hackathon**
Weightage:	**30% of CIA**	**Total Marks:**	**100**
CO4 Statement:	**To understand the concept of supervised and unsupervised methods in machine learning**		
Student Name:	**A. Reddy Prasad**	**Reg. No.:**	**192472045**
Batch	**2024**	**Date:**	**29-08-2026**

Code of Conduct

I A. Reddy Prasad (192472045) **(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.**

Signature of the Student: A. Reddy Prasad

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

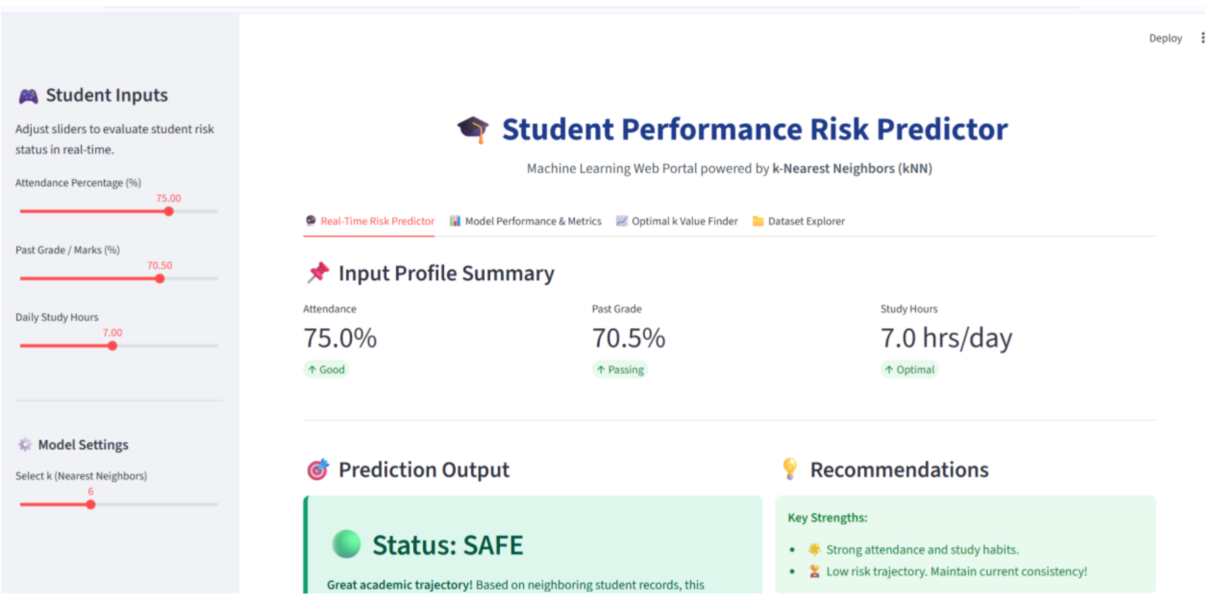
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

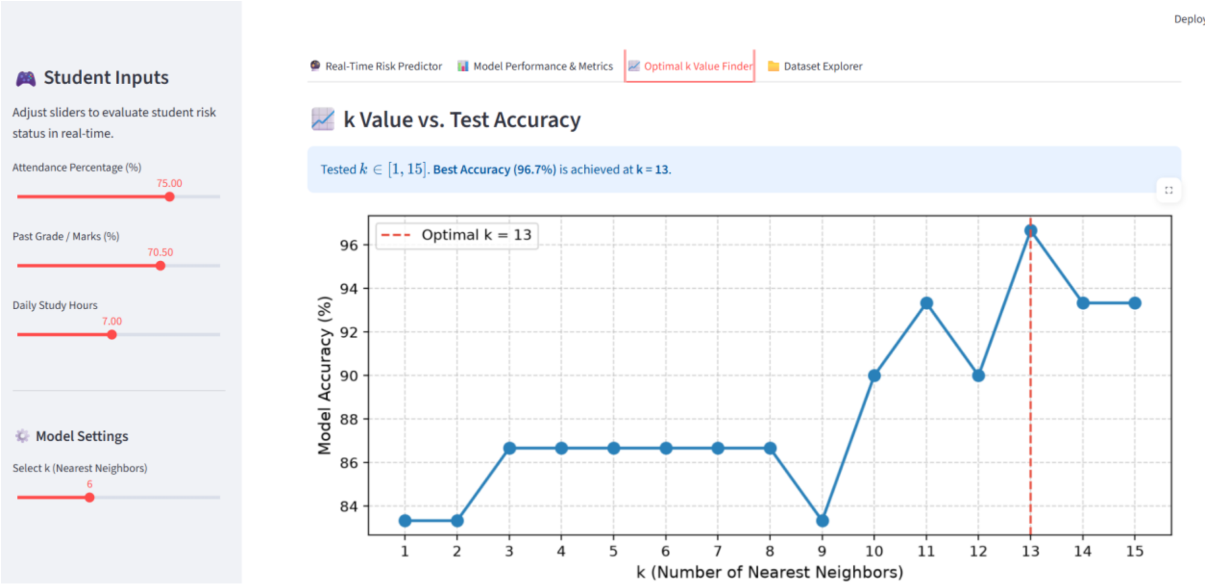
Problem Statement:

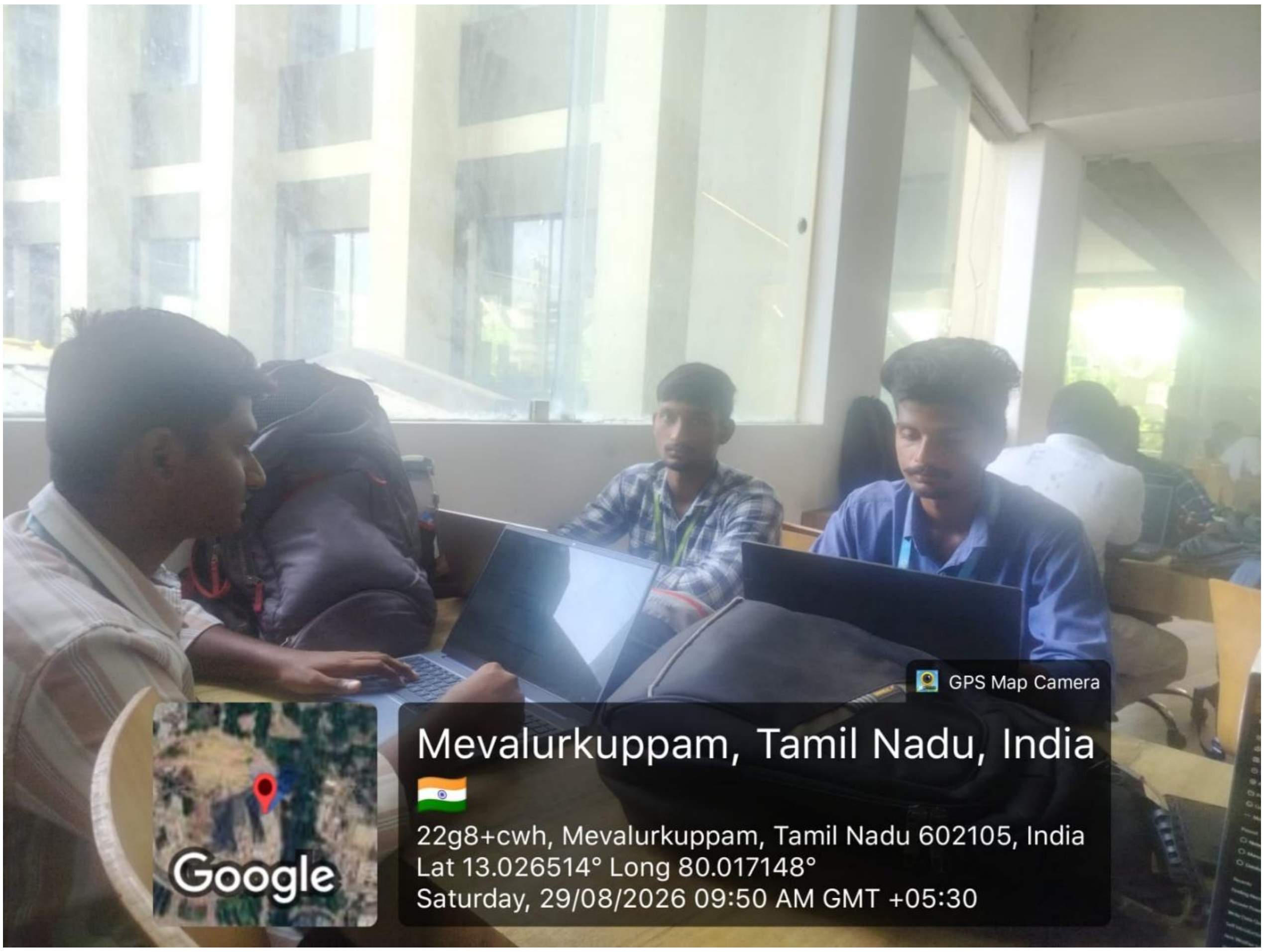
Students at academic risk are often identified too late for effective intervention. This system uses attendance, past grades, and study hours with the kNN algorithm to classify students as At Risk or Safe, helping educators identify at-risk students early and provide timely academic support.

Code:

```
E: > survillence > student.py > ...

import sys
import subprocess

# -------------------------------------------------------------------
# STEP 0: AUTO-INSTALL MISSING LIBRARIES & AUTO-LAUNCH WEB APP
# -------------------------------------------------------------------
required_packages = {
    "streamlit": "streamlit",
    "pandas": "pandas",
    "numpy": "numpy",
    "sklearn": "scikit-learn",
    "matplotlib": "matplotlib",
    "seaborn": "seaborn"
}

for module_name, pip_name in required_packages.items():
    try:
        __import__(module_name)
    except ImportError:
        print(f"[INFO] Installing missing library: '{pip_name}'...")
        subprocess.check_call([sys.executable, "-m", "pip", "install", pip_name])

import streamlit as st

# Auto-launch Streamlit web server if executed via standard 'python student.py'
if not st.runtime.exists():
    print("\n" + "="*60)
    print("LAUNCHING STUDENT RISK PREDICTOR WEBSITE IN YOUR BROWSER...")
    print("="*60 + "\n")
    subprocess.run([sys.executable, "-m", "streamlit", "run", sys.argv[0]])
    sys.exit(0)

# Standard imports once inside Streamlit runtime
import os
import warnings
```




Deploy
Student Inputs
Adjust sliders to evaluate student risk status in real-time.
Attendance Percentage (%)
75.00
Past Grade / Marks (%)
70.50
Daily Study Hours
7.00
Model Settings
Select k (Nearest Neighbors)
6
Real-Time Risk Predictor
Model Performance & Metrics
Optimal k Value Finder
Dataset Explorer
k Value vs. Test Accuracy
Tested $k \in [1, 15]$. Best Accuracy (96.7%) is achieved at k = 13.
Optimal k = 13
Model Accuracy (%)
k (Number of Nearest Neighbors)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